Peilin **Wu**

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Education

SJTU (Shanghai Jiao Tong University) *Shanghai, China*

B.E. in Computer Science and Technology *Sep. 2022 - Present*

- IEEE Honors Class
- GPA: 90.5/100 (3.93/4.3)
- Relevant Coursework and Grades: Mathematics Analysis (90/100) · Linear Algebra (94/100) · Probability and Statistics (94/100) · Linear and Convex Optimization (96/100) · Information Theory (93/100) · Discrete Mathematics (91/100) · Robotics (96/100) · Operation System (95/100)

Publications

4D Latent World Model for Robot Planning

Zhiyi Li, **Peilin Wu**, Xiaoshen Han, Yilun Du *Jul. 2025 - Oct. 2025*

Preprint

- Built a 4D latent world model predicting future 3D structures conditioned on current observations and text goal, achieving high visual quality, physical consistency, and robust viewpoint generalization.
- Developed a planning framework which leverages our model's detailed 3D predictions as geometrically rich goals for an inverse dynamics controller, empowering precise and spatially aware manipulation.

LoopSR: Looping Sim-and-Real for Lifelong Policy Adaptation of Legged Robots

Peilin Wu, Weiji Xie, Jiahang Cao, Hang Lai, Weinan Zhang *Jun. 2024 - Mar. 2025*

***IROS*, 2025**

- Modeled the lifelong learning problem and proposed methods to facilitate a simulated reconstruction of the real world.
- Designed a lifelong policy adaptation framework enhancing performance by at least 30% in the most difficult cases compared to sim-to-real transfer baselines and successfully addressed problems such as catastrophic forgetting.

Bridging the Sim-to-Real Gap from the Information Bottleneck Perspective

Haoran He, **Peilin Wu**, Chenjia Bai, Hang Lai, Lingxiao Wang, Ling Pan, Xiaolin Hu, Weinan Zhang *Apr. - Jun. 2024*

***CoRL* (Oral), 2024**

- Provided theoretical analyses to model the sim-to-real gap with respect to privileged information and historical trajectories.
- •Proposed an efficient and effective sim-to-real transfer method inspired by information bottlenecks, which outperformed existing baselines such as DreamWaQ and RMA by about 10% in simulated RL tasks and real-world quadruped locomotion.

Reasearch Experience

Harvard University *Boston, U.S.*

Research Intern at Harvard Embodied Mind Lab, Advised by Prof. Yilun Du *Jul. 2025 - Present*

Research Topic: World Models, Robotics, Multimodal Perception

Shanghai AI Lab *Shanghai, China*

Student Intern at Embodied AI Center, Advised by Dr. Jingbo Wang *Mar. 2025 - Present*

Research Topic: Human Motion Generation Models, Humanoid Robots

Shanghai Jiao Tong University *Shanghai, China*

Research Assistant at APEX Lab, Advised by Prof. Weinan Zhang *Sep. 2023 - Present*

Research Topic: Quadruped Robots, Reinforcement Learning

Shanghai Jiao Tong University *Shanghai, China*

Research Assistant at MAGIC Lab, Advised by Prof. Siheng Chen *Jul. 2023 - Mar. 2024*

Research Topic: Drone System, Collaborative Communication

Highlighted Projects

Tactile Benchmark: Simulated Visual-Tactile Benchmark for Robotics Manipulation *Boston, U.S.*

Research Assistant at Harvard Embodied Mind Lab, Advised by Prof. Yilun Du *July. 2025 - Present*

- Established an open-source benchmark for tactile sensors across different simulators, including different simulators and tasks.
- Implemented reinforcement learning and imitation learning algorithms.

World Model for Robot Manipulation with Tactile Information *Boston, U.S.*

Research Assistant at Harvard Embodied Mind Lab, Advised by Prof. Yilun Du *July. 2025 - Present*

- Constructed a multimodal world model to generate both visual and tactile information through diffusion-based techniques.
- Developed policies for conducting contact-rich robot manipulation tasks under the supervision of the world model.

Bridging the Gap between Human Motion Generation and Humanoid Control
Shanghai, China

Research Assistant at Embodied AI Center, Advised by Dr. Jingbo Wang
Mar. 2025 - Present

- •Established a thorough pipeline from text or goal-conditioned motion generation to low-level locomotion of humanoid robots.
- •Implemented RL-based fine-tuning techniques to generate a robust system with the ability to continuously improve.

Drone System Construct and Communication for UAV swarm
Shanghai, China

Research Assistant at MAGIC Lab, Advised by Prof. Siheng Chen
Aug. 2023 - Mar. 2024

- Constructed a drone system based on ROS, carrying a GPS sensor and USB camera, leveraged to collect data for autonomous driving datasets.
- Implemented the communication for UAV swarm based on TCP/UDP in preparation for future research on collaborative communication.

Vision System for 6-DoF Robot Arm
Shenzhen, China

Team Member at SJTU RoboMaster Team in RoboMaster Competition 2023
Feb. - Aug. 2023

- Designed algorithms to identify the camera pose and desired end effector pose from RGB image.
- Constructed a thorough pipeline to connect the sensor, PC, and lower computer based on ROS and serial communication.

Honors & Awards

2023 **1st Prize**, China University Robot Competition RoboMaster

2022-2024 **Zhiyuan Honorary Scholarship**, top 5 % of students in SJTU

2023, 2024 **University Scholarship**, top 5 % of students in SJTU

Skills

Programming Python, C/C++, LaTeX, MATLAB, HTML, CSS, JavaScript

Frameworks PyTorch, Tensorflow, NumPy, OpenCV, ROS, Flask

Language Chinese, English (TOEFL 112, GRE 328+3.0)